

7 April 2026

My focus was to have the AI generate quizzes with feedback after submission. The quiz submission flow already existed but had no AI involvement after a student finished. I upgraded it so that when a student submits, their score and the questions they got wrong are sent to Groq, which writes a short personalised feedback message that adjusts its response based on the score, points out what to review.

I then added a Generate a Practice Quiz panel at the top of the Quizzes page where any student can type any topic, pick a difficulty and question count, and instantly get a Groq generated quiz to take privately without it being saved anywhere. Along the way I hit a DTO validation bug on /ai/generate quiz. All the inline DTO classes in the controller were not being used for validation, so all three fields topic, numQuestions, difficulty were being rejected. I moved all DTOs into the shared dto file and updated the controller to import and use them. There is still one small remaining issue where the text in the generator panel inputs appears white in some colour mode. This is partially fixed but needs a final check.

10 April 2026

I planned to build a student dashboard AI insights section that reads each student's real quiz history and attendance data from the database, sends it to Groq, and displays a simple English summary telling the student how they are doing, which subjects they are struggling with, and what to focus on next. I added adaptive difficulty: if a student scores low on a quiz, the platform should automatically suggest a self test on the same topic at an easier level rather than just moving on. I added a one click Generate AI Report button inside the analytics dashboard that takes the class's attendance and quiz performance data and asks Groq to write a readable summary a teacher could actually act on. This has a couple issues with the backend code I am still working out.